knows the worth of a HUD. Relaying critical flight data in the pilot's field of vision not only makes flight stress free, but it reduces information saturation/overload and can save precious seconds during emergency manoeuvres.

Modern automotive HUD implementations are poised to do just that, and more, by incorporating the latest in augmented reality (AR) technology.

While most top auto manufacturers have implemented some basic form of an HUD that's generally restricted to relaying critical information, BMW has been working hard towards imbuing HUDs with the limitless possibilities of AR technology.

The German auto giant has been busy testing prototypes that overlay information on top of what the driver can see through the windscreen. This includes real-time visual warnings for drivers who are in danger of hitting obstacles and other vehicles.

Throw a GPS into the mix and you're looking at turn-by-turn navigation relayed right onto your windscreen. What you have is a GPS system that colour codes and highlights entire lanes to create a breadcrumb trail-based GPS navigation system that's simple enough to be followed by your grandparents.

Other implementations employ a smart array of optical sensors to detect road hazards and enhance their outlines through the HUD. This could prevent accidents even in low visibility conditions.

While that may seem like a glorified Windows Accessibility Option for cars, it still could potentially save lives. BMW is also working on another project that leverages AR technology for service technicians.

The company's prototype AR glasses allow them to diagnose problems and receive step-by-step instruction for fixing the same, just by looking at a malfunctioning car component. However, if you ask me, a much simpler solution would be to hire qualified technicians who don't need an overpriced digital equivalent of 'Fixing Cars for Dummies'.

BMW's far eastern counterpart, Toyota, has also been showing interest in AR HUDs. The company has gone a step further to redefine domain of the technology from mere utility to leisure. The Japanese approach extends this concept to benefit the passengers. The company has produced working prototypes of HUDs integrated into passenger windows, which allow them to select, identify, and zoom in on objects in the scenery. The best part is that, unlike the driver, passengers can't be distracted into a ditch. This allows the integration of touchscreen functionality onto the AR windscreens.

However, that's going to look terribly rude when pedestrians wonder why you've been frantically pointing at them from the passenger seat for no apparent reason. Then again, it won't be a big deal in Japan, where things aren't considered weird until tentacles are involved.

3D-printed cars are poised to get cheaper.

3D-Printed Cars

Most people's idea of 3D printing is restricted to a simplistic version where they believe it's a means to make "real" 3D objects appear out of thin air. While that's technically correct, it still fails to convey the game-changing potential of the technology with respect to the auto industry.

To fully understand its impact, it's best to compare 3D printing with its older cousin – CNC machining. The latter is a subtractive process wherein material is removed to create 3D objects. This fabrication method cannot produce certain complicated shapes, unlike an additive process such as 3D printing. That's precisely why supercar maker Koenigsegg chose to 3D print the turbocharger for its flagship One:1 hypercar, which resulted in significantly improved performance.

However, precision isn't the only criterion going for 3D printing technology. An additive process is inherently more efficient at using raw materials, unlike CNC machining where a great deal of material is wasted.

This allows the manufacturing process to be scaled to an unprecedented extent. US-based Local Motors is a fine example the technology's game-changing potential. It recently built a working car right on the floor of the 2015 North American International Auto Show to prove a point. Even if you exclude the manufacturing costs, the manufacturer estimates that its concept of 3D printing-enabled "micro-factories" result in a whopping 97 percent savings in auto freight and distribution costs alone in relation to what a typical auto manufacturer will pay.

This isn't restricted to auto companies. Further advancements and cost reduction in the technology could herald a future where you could be downloading car parts and modifications right off the internet to print out at your local 3D printer, or possibly even from the comfort of your own house. However, if video game mods available on the internet are any indication, brace yourself for a future where you could have the misfortune of witnessing phallic symbolism on our "clean" roads.